

G Cosmos

An 8-voice analogue pad synthesizer in the spirit of the classic 80s polys, with a detuned layer, chorus and an ambient delay.



Overview

G Cosmos is a polyphonic pad synth in the spirit of the classic 80s polys, with a nod to the layered shimmer of the late-80s digital synths. One oscillator per voice gives saw, pulse (with PWM), a sub-octave and noise; a second, detuned saw (the LAYER) can sit at the same pitch or an octave either side for width and shimmer.

The voices go through a 24 dB filter with its own envelope, a fixed high-pass with four settings, and an amp envelope. The famous bucket-brigade stereo chorus (I, II or both) and an ambient stereo delay with diffused, gently moving repeats finish the sound.

DRIFT makes it behave like real analogue hardware: the tuning and filter wander for the whole instrument and for each voice on its own, and every note lands slightly differently in tune.

Reach for it for

- Lush, slow-moving pads and drones
- Warm 80s poly chords and string machines
- PWM and choir-like textures
- Filter-sweep pads for builds and intros

- Ambient beds with long, diffused delay tails

Playing it

G Cosmos is an instrument: it has no audio input and makes its sound from MIDI.

1. Insert G Cosmos on a track (as the first plug-in, or after a MIDI effect).
2. Arm the track for recording and set its input to your MIDI keyboard (or draw notes in a MIDI item).
3. Pick a patch with < and >, or click the patch display for the list.
4. Play. The MIDI light flashes on each note, and the display shows how many of the 8 voices are sounding.
5. If you use a multi-channel MIDI setup, set CH in the PERFORMANCE section to the channel you want; OMNI listens to all of them.

G Cosmos responds to pitch bend (+/-2 semitones), the mod wheel (CC1 adds vibrato through the LFO, up to 40 cents), the sustain pedal (CC64) and all-notes-off (CC120 and CC123).

Controls

The header

Patch display	Shows the patch number and name, how many of the 8 voices are playing and the chorus setting. Click it to open the patch list.
< and >	Step to the previous or next patch. A patch sets every sound control and VELOCITY; DRIFT, VOLUME and the MIDI channel are left as they are. Loading a patch stops any notes still sounding.
MIDI	Lights when a note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.

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Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

LFO

RATE	LFO speed, 0.05 - 20 Hz.
DELAY	How long the LFO takes to fade in, 0 - 5 s. It starts again from the first key of each new phrase (when no other keys are held), so held chords grow vibrato and movement as they sustain.
SHAPE	SINE, TRI, SQUARE or RANDOM (stepped).
PITCH	Vibrato depth, 0 - 100 cents. The mod wheel adds up to 40 cents more.

The LFO also moves the filter (LFO in the VCF section) and the pulse width (PWM in the DCO section).

DCO

SAW	Level of the sawtooth, 0 - 100%.
PULSE	Level of the pulse wave, 0 - 100%.
PW	The pulse width, 5 - 95%. 50% is a square; narrower settings sound thinner and more nasal.
PWM	How much the LFO sweeps the pulse width, for the classic moving, chorused pulse sound.
SUB	Level of a square wave one octave below.
NOISE	Level of white noise, for breath and air.

LAYER

LEVEL	Level of the second, detuned saw, 0 - 100%.
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DETUNE	How far the layer is detuned, 0 - 50 cents. Small amounts give a slow beating width; large amounts a thick, wide shimmer.
OCTAVE	The layer's octave: -1, 0 or +1.

HPF

0 / 1 / 2 / 3	The fixed high-pass after the filter. 0 BOOST lifts the bass below about 120 Hz; 1 FLAT leaves it alone; 2 and 3 THIN cut the lows (from about 240 Hz and 720 Hz), so the pad sits above a bass part.
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VCF

CUTOFF	The 24 dB low-pass filter's cutoff, 20 Hz - 18 kHz.
RES	Resonance, 0 - 100%. High settings ring and sharpen the sweep.
ENV	How far the FILTER ENV moves the cutoff, -100 to +100%. Negative values sweep it down instead of up.
LFO	How much the LFO moves the cutoff, 0 - 100%.
KEY	Key tracking, 0 - 100%: how much higher notes open the filter. At 100% the cutoff follows the keyboard.

FILTER ENV and AMP ENV

Two separate ADSR envelopes: FILTER ENV moves the cutoff (by the ENV amount), AMP ENV shapes the volume.

ATTACK	Time to rise, 1 ms - 8 s.
DECAY	Time to fall to the sustain level, 5 ms - 10 s.
SUSTAIN	The level held while the key is down, 0 - 100%.
RELEASE	Time to die away after the key is let go, 5 ms - 15 s.

CHORUS

I / II

The bucket-brigade stereo chorus. I is a slow, gentle sweep; II is faster and deeper. Light both for I+II, where the two sweeps run together. With both off the chorus is out.

DELAY

An ambient stereo delay: the repeats bounce between left and right (the right side at three-quarters of the time), are diffused, move gently and get darker as they fade.

TIME	Delay time, 40 - 1500 ms.
FEEDBACK	How many repeats, 0 - 95%.
STONE	How dark the repeats are, 500 Hz - 12 kHz.
MIX	How loud the delay is, 0 - 100%.

PERFORMANCE

VELOCITY	How much playing harder makes notes louder, 0 - 100%. At 0% every note plays at the same level.
DRIFT	See Drift below.
VOLUME	Output level, -24 to +6 dB.
CH	The MIDI channel G Cosmos listens to: OMNI (all) or CH 1 to CH 16.

The output is softly clipped at -0.3 dBFS, so peaks never go over.

Drift

DRIFT

0 - 100% (25% by default). The tuning wanders slowly for the whole instrument (up to 20 cents at 100%) and for each voice on its own (up to 6 cents); the filter cutoff wanders the same way, for the whole instrument and per voice; the layer's detune drifts a little; and every note starts slightly out of tune (up to 6 cents). At 0% everything stays exact.

Presets

Cosmos	The starting sound: saw and sub with a detuned layer, a slow swell, chorus I and an ambient delay.
Warm Poly	Saw and sub with the bass BOOST, a darker filter, a quicker attack and chorus II.
Strings 6	Bright saws with an octave-up layer, a thinner HPF, gentle vibrato and both choruses.
Glass Shimmer	A strong, widely detuned layer an octave up, a bright filter, long release and a long, wet delay.
Dark Drone	Bass BOOST, a low, resonant filter with slow LFO movement, a sub-octave layer, very slow attack and release and a long, dark delay.
PWM Haze	A pulse wave swept by the LFO (PWM) with sub, over chorus I.
Brass Pad	Punchy saws with a strong filter envelope, a fast attack, light vibrato and more velocity.
Choir Air	A narrow pulse with PWM, saw, noise and a layer, resonant and thin, with vibrato and both choruses.
Sweep Pad	A low, resonant filter that slowly opens on each chord (slow filter attack and decay).
Slow Strings	Saws with a strong layer, a very slow attack, vibrato and both choruses.
Nebula	Everything: saw, PWM pulse, sub, noise and a wide octave-up layer, a random LFO on the filter, very long envelopes and a long, heavy delay.
Init	The default settings (the same as Cosmos).

Tips

- For pads that bloom, set the LFO DELAY to 1 - 2 s and add a little PITCH and filter LFO: each new chord starts still and grows movement.
- In a busy mix, use HPF 2 or 3 so the pad stays out of the way of the bass.
- A negative ENV with a high CUTOFF gives pads that darken as they sustain.
- Chorus I+II with a little layer DETUNE is the widest setting; try it on a single held note before adding more voices.
- Use VELOCITY at 0% for even sequenced pads, and higher for played parts.

File: gfx-g-cosmos.jsfx